AI-Powered Travel Planner Agent

IBM SkillsBuild University Internship Project Documentation | Domain: Travel, Tourism & Agentic AI

> **Project Overview**
> The AI-Powered Travel Planner Agent is an intelligent multi-agent assistant built to streamline trip planning, itinerary creation, and localized destination discovery. Leveraging IBM watsonx.ai, IBM Granite models, and Langflow, the system coordinates specialized agents to deliver personalized travel insights and dynamic itineraries.

1. Problem Statement & Objectives

The Challenge: Planning trips efficiently is complex due to scattered information across multiple platforms regarding destinations, transport, weather, and local attractions. Travelers often struggle to create optimal itineraries or manage trip budgets without tailored, real-time guidance.
The Objective: Build an AI-powered multi-agent system that delivers personalized travel insights, dynamic itineraries, and real-time guidance through structured modular components.

2. Proposed Solution & Architecture

The solution utilizes a Retrieval-Augmented Generation (RAG) pipeline to fetch accurate travel data from trusted datasets. This data is embedded and stored in a vector database, enabling the system to retrieve and summarize destination details using IBM Granite models.

- **City Selection Agent:** Evaluates user criteria, budget constraints, and geographic destination parameters.
- **Local Expert Agent:** Gathers location-specific insights, cultural recommendations, and guidelines via RAG.
- **Travel Concierge Agent:** Synthesizes inputs into a coherent, optimized day-by-day itinerary.

3. Technology Stack

Component	Technology / Service Used
Workflow Orchestration	Langflow platform for visual multi-agent workflow design.
Generative AI Models	IBM Granite Models (e.g., `granite-4-h-small`) hosted on IBM watsonx.ai.
Cloud Infrastructure	IBM Cloud for scalable, secure, and reliable deployment.
Knowledge Retrieval	RAG pipeline with Vector Database (FAISS/Chroma) and Guardrails.